Simulation of High-Efficiency FTO/ZnSe/CdTe/Cs2BiAgI6 /Ni Solar Cells Using SCAPS-1D

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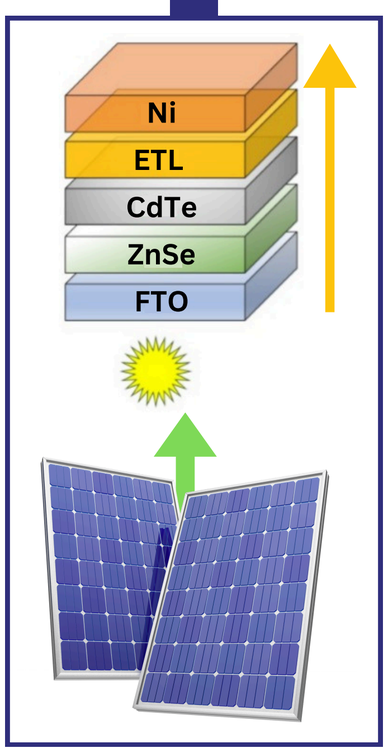
Introduction

To meet the growing energy demand worldwide, alternative sources like solar energy are crucial. Thin-Film Solar Cells (TFSCs) using novel materials have emerged as a viable alternative to silicon wafers. This study utilizes a relatively unexplored combination of CdTe (Cadmium Telluride) and ZnSe (Zinc Selenide). CdTe has shown promise due to its excellent optoelectronic properties, while ZnSe, a metal chalcogenide, is a novel alternative to the more common, and toxic CdS. This investigation aims to optimize the given configuration by measuring performance metrics, namely Open Circuit Voltage, Current Density, Fill Factor, and Power Conversion Efficiency using SCAPS-1D under standard illumination conditions.



Objective

To investigate the optimal intrinsic parameters (Temperature, Wavelength, etc.) of different solar cell layers and their effect on solar cell performance.

To determine the optimal length to enhance the quantum efficiency of the given solar cell.

To investigate the performance parameters of solar cells, viz. V (Open Circuit Voltage), J (Current Density) & FF (fill factor).

Methodology

This study utilizes the SCAPS-1D (Solar Cell Capacitance Simulator in 1 Dimension) software. SCAPS-1D is a simulation program developed by Professor Marc Burgelmann. SCAPS-1D provides the flexibility to calibrate up to 7 layers and add many intrinsic parameters, such as thickness, energy bandgap, electron affinity, relative permittivity, defects, metalwork functions, and illumination conditions. This tool implements Poisson's equation and the semiconductor continuity equations simultaneously.

$$\frac{d^2}{dx^2}\psi(x) = \frac{q}{\varepsilon_r}\left[p(x) - n(x) + N_D - N_A + N_t\right]$$

$$\frac{1}{q}\frac{d}{dx}J_n = R_n(x) - G_{OP}(x)$$

$$\frac{1}{q}\frac{d}{dx}J_p = G_{OP}(x) - R_n(x)$$

$$V_{bi} = E_{g2|} - E_{g1}$$

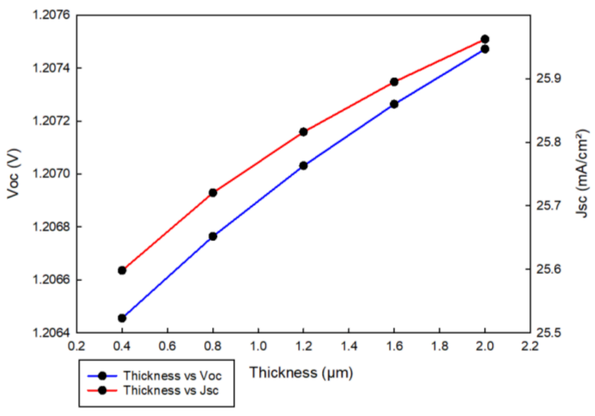
$$J_n(x) = q\mu_n E(x) + qD_n\nabla_n(x)$$

$$J_p(x) = q\mu_p E(x) + qD_p\nabla_p(x)$$

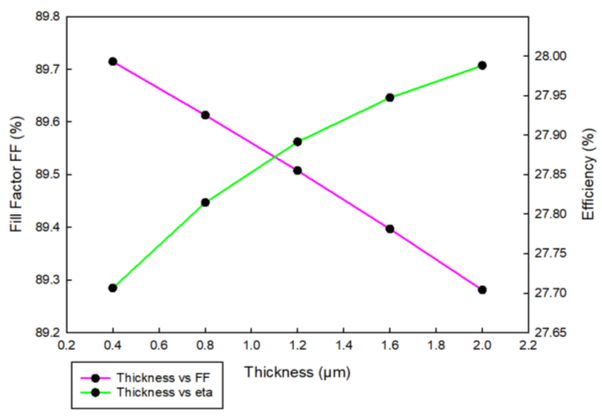
$$J = J_{ph} - J_0[e^{(qV/aK_bT)} - 1$$

Results & Discussion

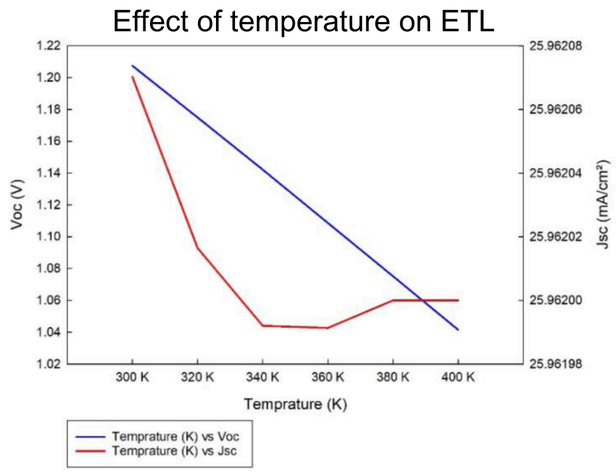
Effect of V and J with thickness on ETL



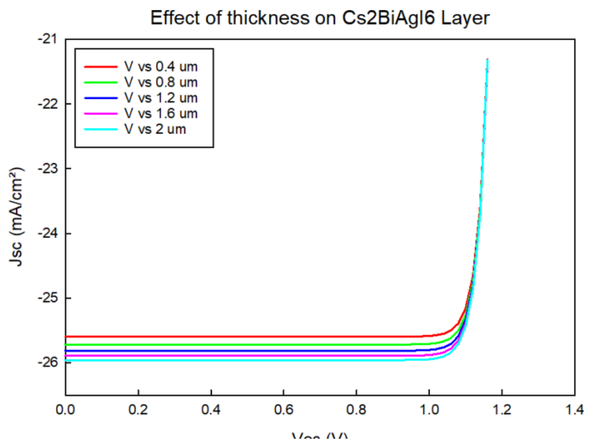
Effect of FF and PCE with thickness on ETL



Effect of temperature on ETL



Effect of thickness on Cs2BiAgI6 Layer



Recommendations

- Explore alternative ETLs like Buckminsterfullerene (C60) which also gave a high PCE.
- Identify optimal thicknesses for each layer based on their band gaps to ensure maximum efficiency
- Investigate other non-refractory buffer layers like ZnSe.

Conclusion

This study employed SCAPS-1D to optimize photovoltaic cell performance, focusing on electron transport layers (ETLs) and buffer materials. Cs_2BiAgI_6 demonstrated excellent potential, achieving a PCE of 27.99%, which increased to 28.45% with selenium as the back contact. Key metrics, including VOC (1.2075 V), JSC (25.962 mA/cm²), and FF (89.28%), were optimized. The study also offers future insights on buffer layers as ZnSe was introduced as an alternative choice, offering improved efficiency and cost-effectiveness, paving the way for high-performing, affordable solar cell designs. It was also found that temperatures beyond ambient conditions decrease the efficiency of the cell, due to a higher degree of random molecular motion, where electron and hole carriers could collide and recombine, thus reducing its performance.

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